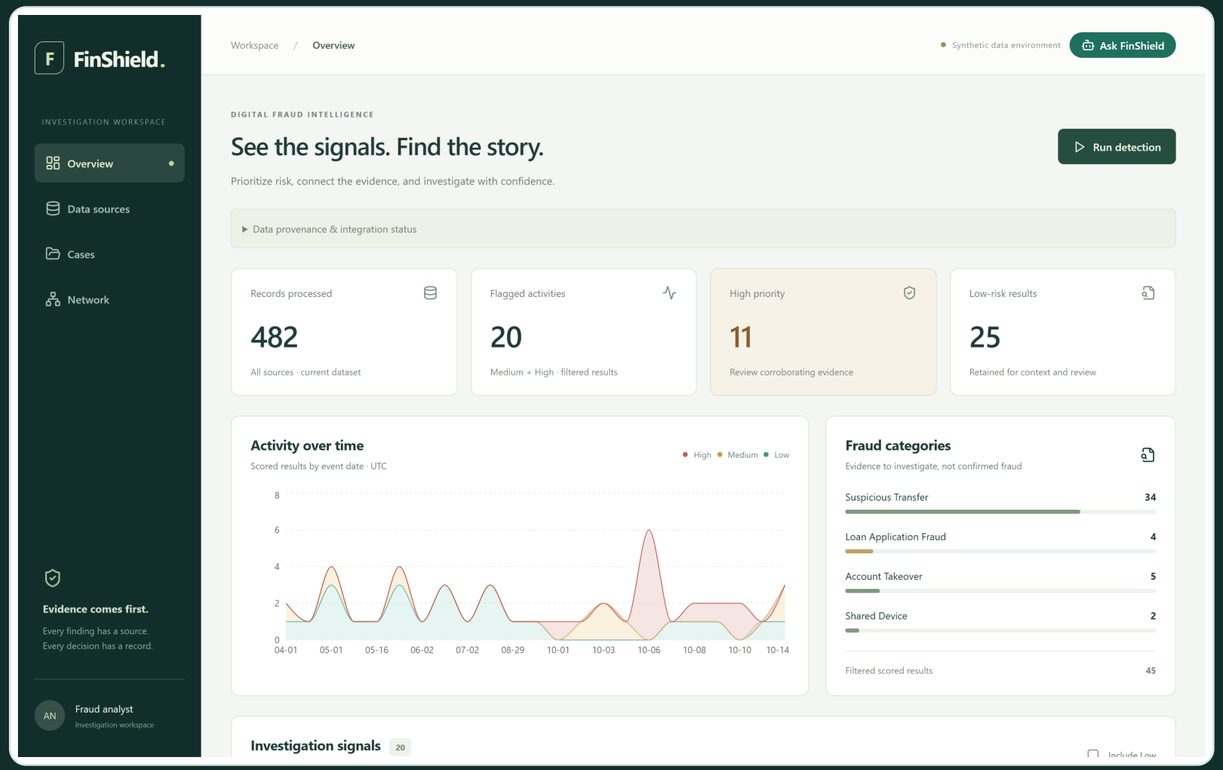
AZENTIO HACKATHON · DIGITAL FRAUD INVESTIGATION SYSTEM

FinShield

Explainable fraud detection and case investigation across banking, lending and insurance data.

Level 1 Ingest, detect, score, explain, dashboard, case queue

Level 2 Correlate across systems, relationship network, consolidated cases, analyst workspace, audit trail and report, prioritisation, AI with guardrails



Team · A: Harsh (data platform, cases, API) · B: mansh7763 (detection, graph, risk) · C: Bhuvan (investigation UI, AI)

All data shown is synthetic. Every number on screen is computed from the ingested records.

OUR APPROACH

Every flag has a reason. Every decision has a record.

An analyst must see **what** was flagged, **why**, on **which records**, and record a decision they can defend later. So FinShield is built as one evidence chain, from file upload to case report:

01 · LEVEL 1

Ingest

CSV, Excel, JSON; validated; rejected rows kept with reasons

›

02 · LEVEL 1

Link

One entity model: customer, account, device, IP, phone, application, claim

›

03 · LEVEL 1

Detect

23 risk indicators + 6 mitigations for legitimate lookalikes

›

04 · LEVEL 1

Score

0 to 100, Low / Medium / High; contributions add up to the score

›

05 · LEVEL 2

Correlate

Alerts grouped into one investigation, with the reason

›

06 · LEVEL 2

Investigate

Network, timeline, actions, AI assessments, audit trail, report

Three principles we never broke

- **Show the why.** Every score is a list of named indicators with the record IDs behind them, including the factors that lowered risk.
- **Nothing silently lost.** Rejected rows, unmapped columns and indicators skipped for missing data are shown with reasons. Missing evidence is never treated as safety.
- **The record is sacred.** Alerts are immutable; decisions, corrections and AI assessments are appended, never overwritten.

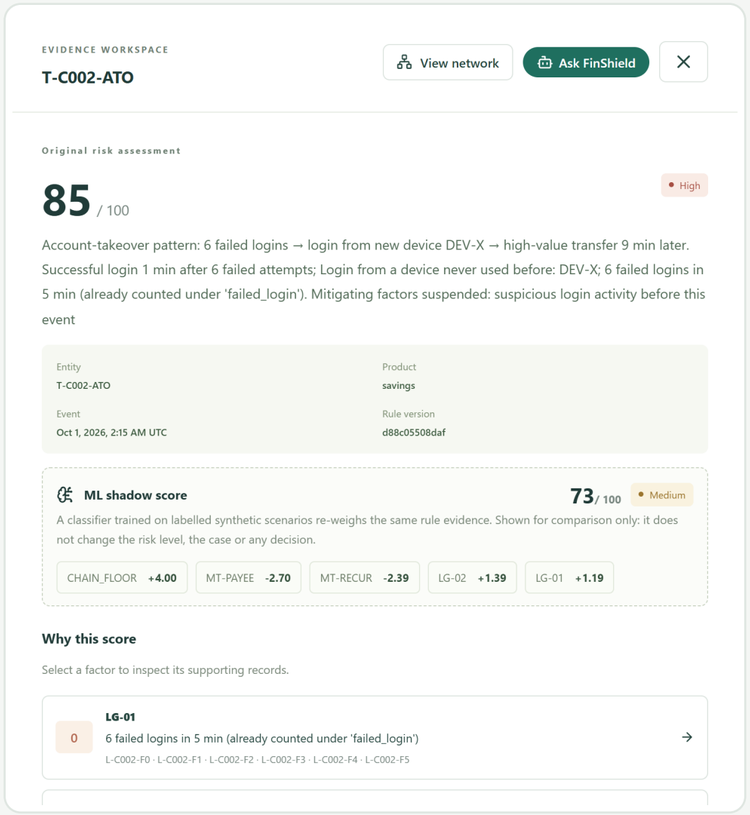
Level 1

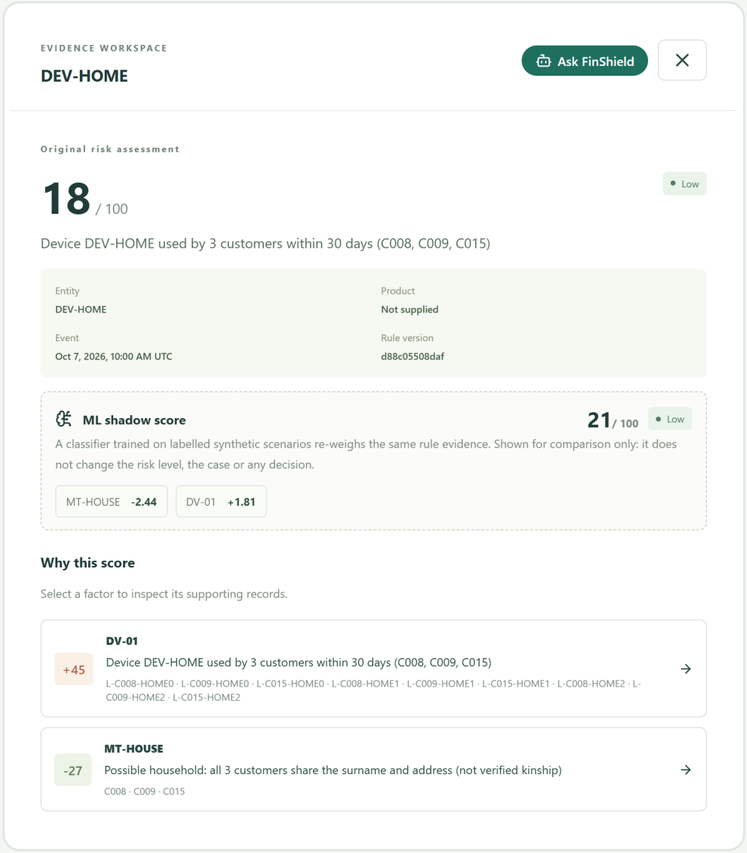
What the brief asked for, and where to see it

Requirement	What we built	Where to see it
Data ingestion and integration	CSV, Excel and JSON upload; header auto-mapping with manual override; mandatory-field validation; rejected rows stored with reasons; duplicates and conflicts reported; records linked by customer, account, application, device and transaction IDs	Data sources tab
Fraud detection (4 required scenarios + extra)	Suspicious transfers, shared devices, repeated login failures and takeover chains, legitimate lookalikes kept Low; extra: suspicious loan applications, loan-to-mule flow, identity reuse, network patterns	Overview → any alert
Fraud risk assessment	Score 0–100 and Low / Medium / High from several independent indicators; customer baselines lower risk for regular behaviour; the takeover chain raises it	Alert: "Why this score"
Fraud alert dashboard	Records processed, flagged events, risk distribution, categories, activity over time; filters for customer, transaction, product, risk level, date and detection type	Overview tab
Basic fraud case queue	Medium and High alerts open cases with ID, customer/account, category, risk, evidence and status; decisions Requires Investigation / False Positive; original alert never changes	Cases tab

Level 1

Same engine, opposite verdicts, and it shows why





Left: account takeover, High 85

6 failed logins → success from a never-seen device → a large transfer 9 minutes later. The chain is corroborated by independent facts, so mitigations are suspended.

Right: family tablet, Low 18

Three customers share one device (+45), but they share a surname and address, so the household mitigation applies (−27). Shown as "possible household", not verified kinship.

What the analyst gets

- Each contribution lists its record IDs; click to see the rows.
- Contributions add up exactly to the score.
- ML shadow score shown beside it for comparison only; it never changes the band or the case.

Level 1

23 risk indicators, 6 mitigations, configurable in one file

Family	Indicators (rules.yaml)	Catches
Transfers	TX-01, TX-03 – TX-09	Amount far above own norm, new payees, velocity, fan-out, rolling totals, structuring just below limits, dormant accounts waking up, mule pass-through
Logins and takeover	LG-01 – LG-03, HV-01, ATO chain floor	Failed-login bursts, success after burst, sensitive action after suspicious login
Devices	DV-01 – DV-03	Device shared by several customers, unrelated customers applying from one device, new device
Applications and identity	AP-01 – AP-04, ID-01, ID-02	Loan swarms on one device or payout account, application velocity, loan-to-mule cash-out, contact details reused across identities
Network	NW-01, NW-02	Unrelated customers converging on one recipient; circular money flows
Mitigations	MT-KNOWN, MT-TENURE, MT-RECUR, MT-PAYEE, MT-HOUSE, MT-OWN	Lower risk for known payees, long tenure, recurring payments, households and own-account sweeps

Every indicator is a weight and parameters in **backend/app/detect/rules.yaml**; a change creates a new rules version and the old alerts stay as history. Ingestion keeps provenance (file, row, raw values) for every record.

EVALUATION

Measured on held-out synthetic data we did not tune on

93%
of 56 planted fraud incidents caught at Medium or High

0%
of 48 legitimate lookalikes raised to High (8% to Medium)

100%
recall on brand-new customers (cold start, 16 incidents)

34/34
hand-built calibration scenarios behave as designed

357
backend tests at the Level 1 freeze

Learned weights (ML shadow model), tested on 3 unseen datasets

Measure	Hand weights	Learned weights
Ranking quality (average precision)	0.78	0.95
Precision of Medium+ alerts	76%	88%
Background false alarms	35	17
Fraud incidents caught at Medium+	89%	89%

Read these numbers honestly

- Synthetic data with planted, labelled scenarios: it measures the rules against what we can simulate, not real-world fraud rates.
- Missed: "slow drain" (small amounts over time) stays below Medium, an open gap.
- The ML model only re-weighs rule evidence and is shown as a shadow score; the explainable rule score stays official.

Sources: evals/reports/d3_heldout.md, golden_scorecard.md, ml_stability.md, stage1_tests.txt

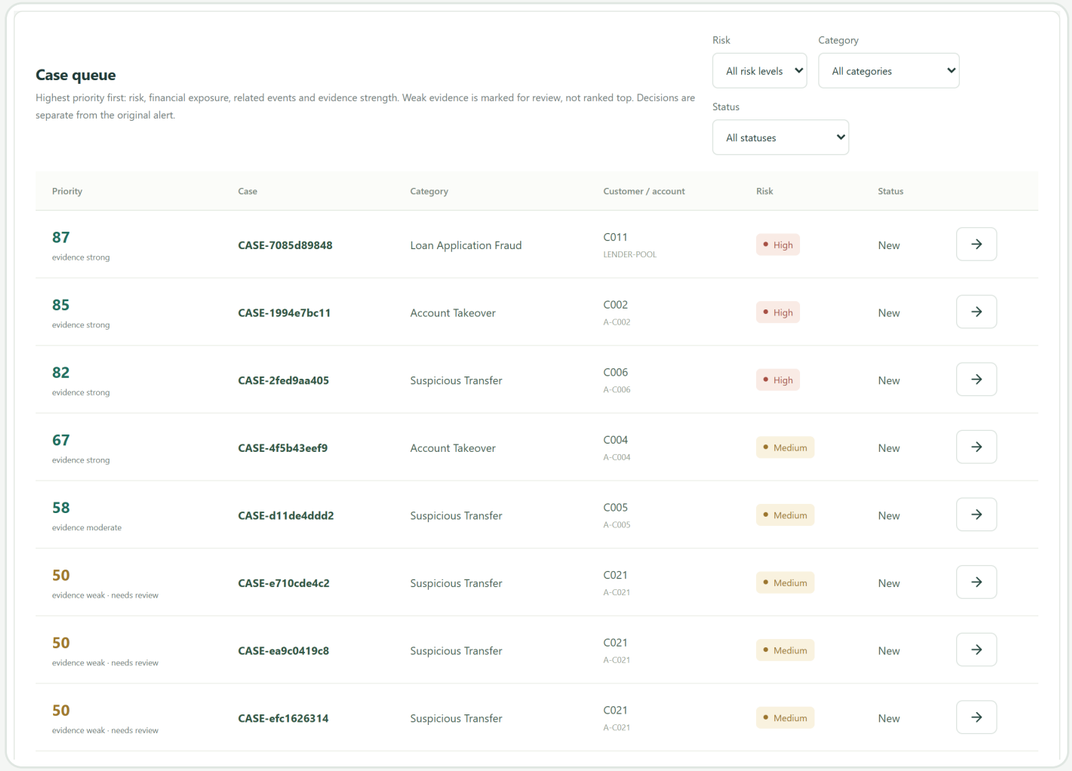
Level 2

Six Level 2 capabilities, all working in the app

#	Requirement	What we built	See it
1	Cross-system correlation	Lending, insurance and banking records linked by shared device, phone, email and payout account (never just a common IP). Loan on a shared device → payout → unfamiliar recipients is one investigation.	Cases → CASE for C011
2	Relationship analysis	Network view; every link shows the shared identifier and the record IDs that prove it; click any entity for its records. Very common hubs are hidden, not invented.	Network tab
3	Case consolidation	Alerts grouped by linked group or same customer within 72 h, with the reason shown; analyst can split wrong groupings; original alerts kept.	Case → Why grouped, Split
4	Investigation workspace	Timeline, entities, indicators, evidence, priority, notes; 8 actions: confirm, false positive, request info, escalate, recommend hold, recommend verification, conclude, reclassify.	Case → Analyst actions
5	Evidence and audit trail	Every decision, action, AI assessment and correction stored with user and time; reclassification keeps the detector's original band; JSON and Markdown case report.	Case → Decision history, Download
6	Prioritisation	Priority from risk, financial exposure (per currency), related events and evidence strength; weak evidence marked "needs review" instead of top priority.	Cases queue

Level 2

The queue tells the analyst where to start

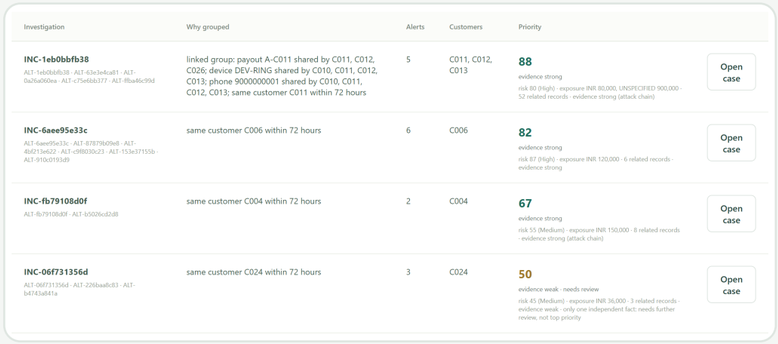


Case queue

Highest priority first: risk, financial exposure, related events and evidence strength. Weak evidence is marked for review, not ranked top. Decisions are separate from the original alert.

Risk: All risk levels | Category: All categories | Status: All statuses

Priority	Case	Category	Customer / account	Risk	Status
87 evidence strong	CASE-7085d89848	Loan Application Fraud	C011 LENDER-POOL	High	New
85 evidence strong	CASE-1994e7bc11	Account Takeover	C002 A-C002	High	New
82 evidence strong	CASE-2fed9aa405	Suspicious Transfer	C006 A-C006	High	New
67 evidence strong	CASE-4f5b43eef9	Account Takeover	C004 A-C004	Medium	New
58 evidence moderate	CASE-d11de4ddd2	Suspicious Transfer	C005 A-C005	Medium	New
50 evidence weak · needs review	CASE-e710cde4c2	Suspicious Transfer	C021 A-C021	Medium	New
50 evidence weak · needs review	CASE-ea9c0419c8	Suspicious Transfer	C021 A-C021	Medium	New
50 evidence weak · needs review	CASE-efc1626314	Suspicious Transfer	C021 A-C021	Medium	New

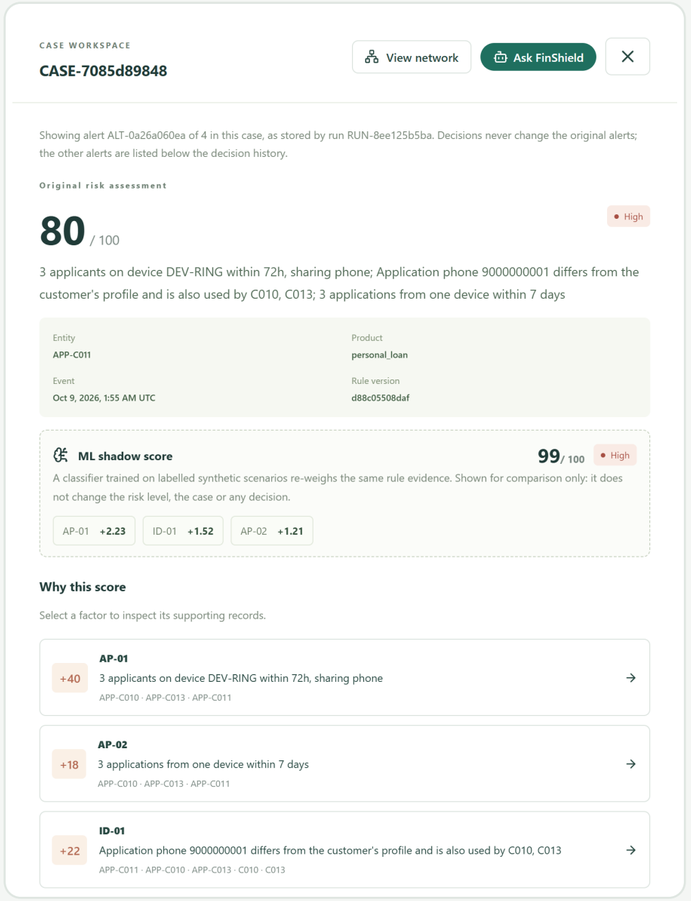


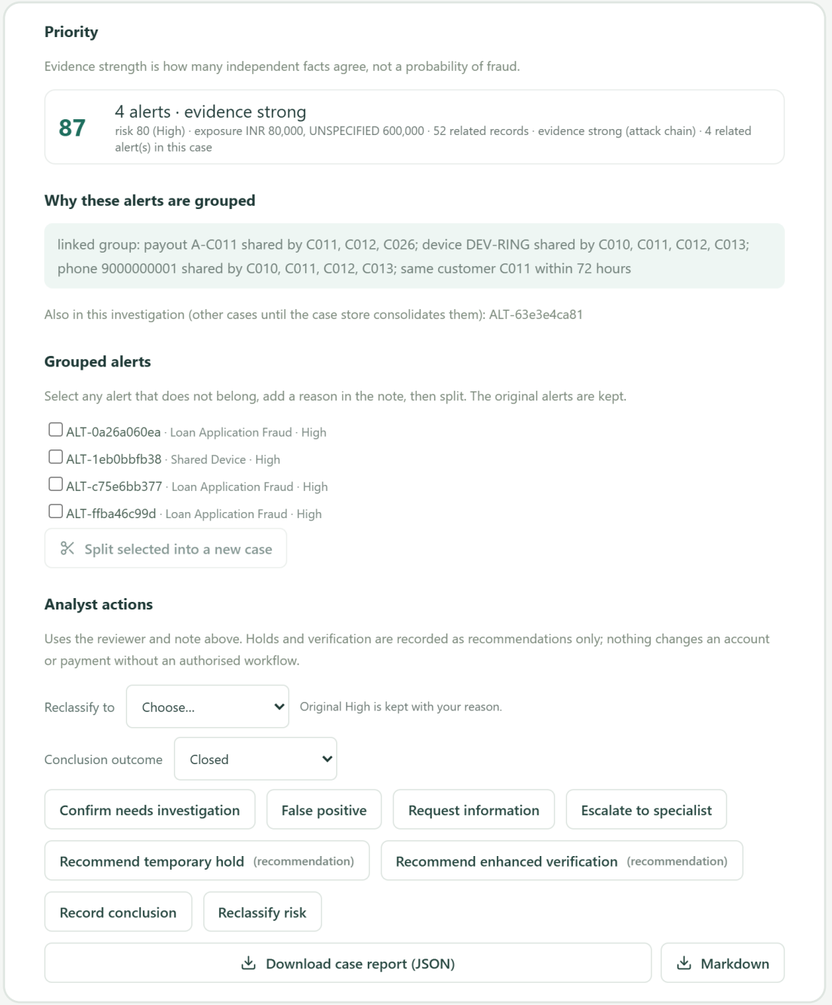
Investigation	Why grouped	Alerts	Customers	Priority	
INC-1eb0bbfb38	linked group: payout A-C011 shared by C011, C012, C026; device DEV-RING shared by C010, C011, C012, C013; phone 9000000001 shared by C010, C011, C012, C013; same customer C011 within 72 hours	5	C011, C012, C013	88 evidence strong	Open case
INC-6aee95e33c	same customer C006 within 72 hours	6	C006	82 evidence strong	Open case
INC-fb79108d0f	same customer C004 within 72 hours	2	C004	67 evidence strong	Open case
INC-06f731356d	same customer C024 within 72 hours	3	C024	50 evidence weak · needs review	Open case

- **Priority** combines risk score, exposure, related events and evidence strength. The C011 loan ring (87, strong) is ranked first.
- **Evidence strength** is strong / moderate / weak: how many independent facts agree. It is not a probability.
- **Needs review:** a case resting on a single fact is capped below the top and labelled, instead of being given the highest risk.
- **Consolidated investigations** list why alerts were grouped, e.g. "device DEV-RING and phone 9000000001 shared by C010–C013; same customer C011 within 72 hours".

Level 2

One screen to examine, decide and document a case

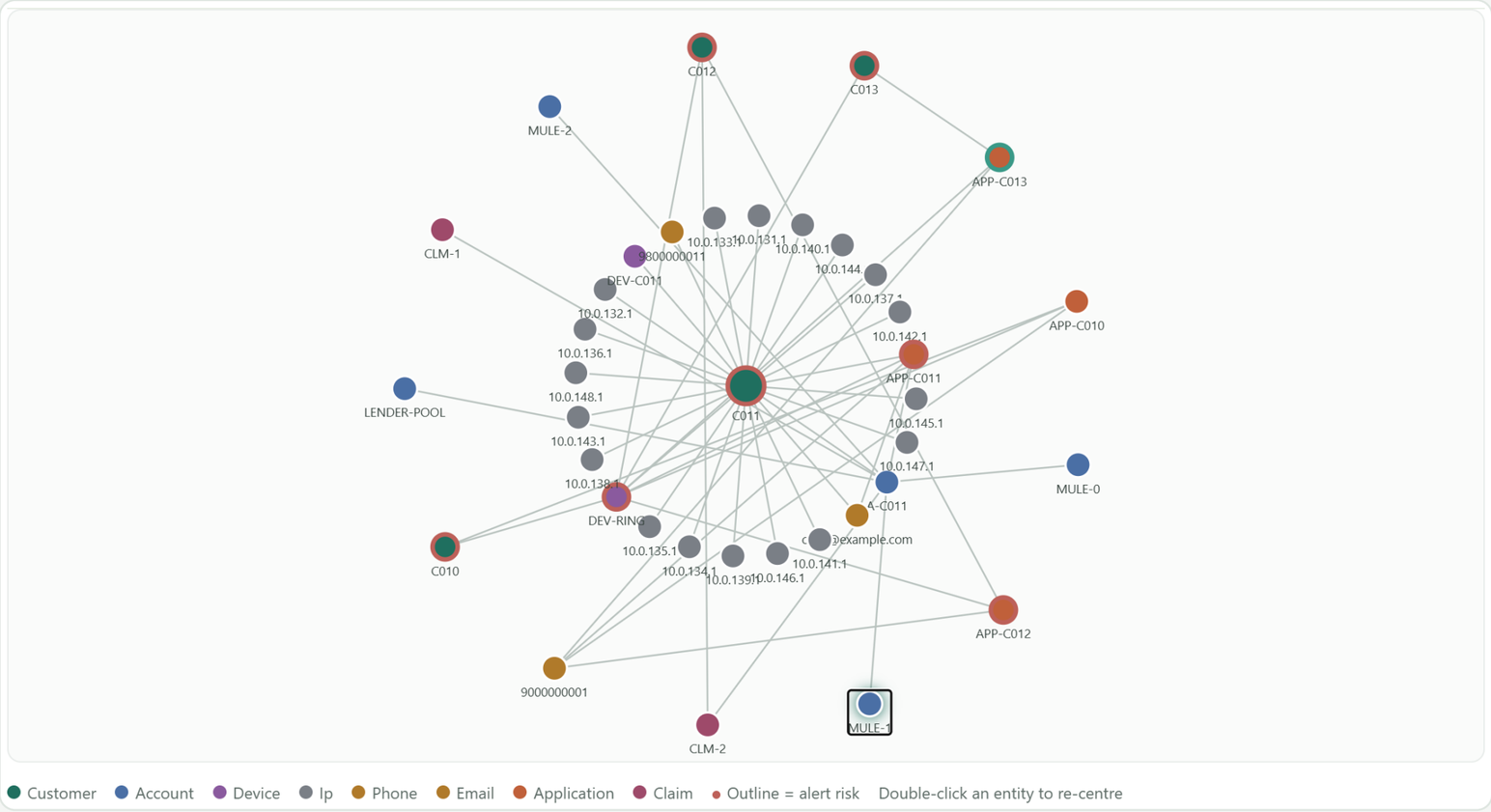




- **Evidence:** original alert snapshot, score breakdown, connected entities, event timeline, source rows.
- **Analyst actions** need a reviewer and a note. Holds and verification are recorded as **recommendations only**: nothing touches an account or payment without an authorised workflow.
- **Reclassify** keeps the detector's original band beside the analyst's.
- **Split** moves wrongly grouped alerts to a new case; both keep the original alerts.
- **Audit trail** of decisions, actions, AI assessments and corrections, with user and time.
- **Download** the structured case report as JSON or Markdown.

Level 2

Relationships come from matching identifiers, not guesses



- Around customer C011: three loan applications on device **DEV-RING** with one phone, payouts into account A-C011, outflows to MULE accounts, and insurance claims CLM-1 / CLM-2 on the same payout account.
- Hover a link to see the shared identifier and its record IDs; click an entity for every record that mentions it.
- Linked groups use device, phone, email or payout account, never just an office IP or a popular payee, so families and colleagues are not treated as rings.

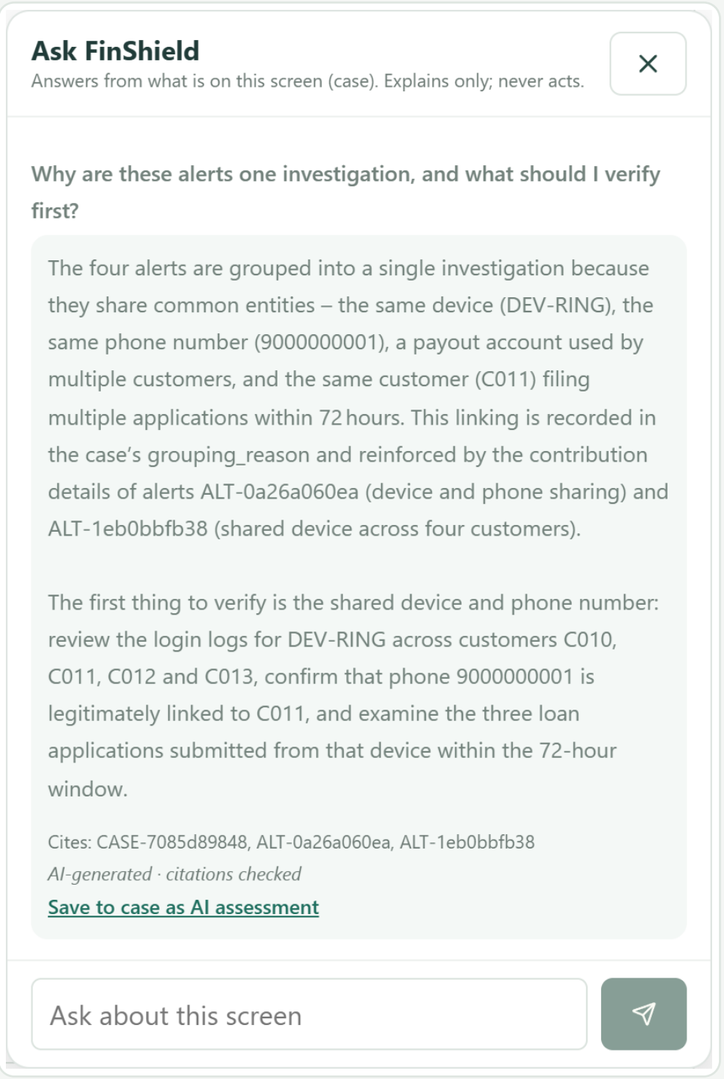
Linked customer groups

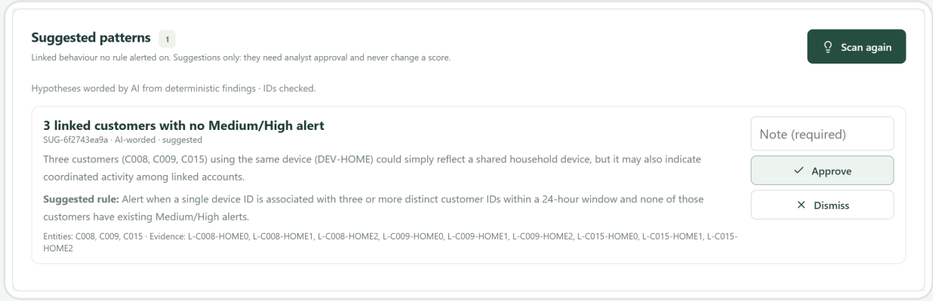
Customers sharing a device, phone, email or payout account (never just an IP or a common payee).

Group	Why linked	Alerts
C008, C009, C015	device DEV-HOME shared by C008, C009, C015	1
C010, C011, C012, C013, C026	payout A-C011 shared by C011, C012, C026; device DEV-RING shared by C010, C011, C012, C013; phone 9000000001 shared by C010, C011, C012, C013	5

Level 2

Two agents, checked against the data





Ask FinShield: questions about what is on screen

- The browser sends only the screen and its IDs; the backend loads that data from the store.
- Every cited ID must exist in that data, otherwise it is dropped and shown as dropped.
- Record text is treated as data, never instructions. Earlier reviewers' notes are included.
- An answer can be saved to the case as an **AI assessment**, labelled, and rejected by the analyst; the original stays in the audit trail.

Pattern Scout: what no rule caught

- Deterministic mining finds linked groups and recipient convergence with no Medium/High alert.
- The LLM only words the hypothesis and a suggested rule; it cannot add entities or change a score.
- Approve or dismiss needs a note; approving never changes the rules.
- Everything works with the LLM switched off (templates, quoted evidence).

Level 2

Three AI agents, one rule: they suggest, the analyst decides

Agent	What it does	How it works	Where to see it
Pattern Scout (workstream B)	Finds linked behaviour that no rule alerted on: customers sharing a device, phone or payout account with no Medium/High alert; several customers starting to pay the same new account.	Deterministic mining finds the candidates and their record IDs. The LLM only words a hypothesis (including an innocent explanation) and a suggested rule. Approve / Dismiss needs a note and is saved; approving never changes rules or scores.	Network tab → Run Pattern Scout
Ask FinShield (workstream C)	Answers questions about the current screen: "Why is this case high risk?", "What links these customers?", "What should I verify first?"	The browser sends only the screen and its IDs; the backend loads that data from the store and gives the LLM nothing else. Cited IDs not in that data are dropped. An answer can be saved to the case as an AI assessment, then rejected by the analyst.	Top-right button on every screen, and inside each case
Reviewer memory (workstreams C + A)	Brings earlier reviewers' decisions, actions and notes on this case, and on cases sharing its customer or account, into the investigation.	Reads the append-only case history; the notes are passed to Ask FinShield as context, never as instructions.	Case → "What earlier reviewers found"

Guardrails shared by all three

Never decide

No agent can change a score, band, case status or account.

Real links only

Relationships and citations must be IDs that exist in the data.

Data, not orders

Text inside customer records is treated as data, never as instructions.

Labelled, rejectable

Saved AI output is marked AI-generated in the audit trail and can be rejected.

Works with AI off

If the LLM is unavailable, the app uses templates and quoted evidence.

Model: openai/gpt-oss-120b on Groq (set by GROQ_MODEL). Not an agent: the **ML shadow score** is a classifier that re-weighs the rule evidence and is shown for comparison only.

What we did not claim, and what comes next

Limits today

- Synthetic data only; metrics are on simulated scenarios, not production fraud rates.
- Reviewer identity is a typed name; there is no login or role-based approval yet.
- "Slow drain" fraud (small amounts over a long time) is not caught at Medium.
- Hold and verification are recommendations; no payment system is connected.
- AI wording depends on Groq; when it is off, the app falls back to quoted evidence.

Next steps

- Analyst feedback (confirmed fraud / false positive) retrains the shadow model.
- Login, roles and four-eyes approval for account-affecting actions.
- Streaming ingestion for real-time scoring.
- Approved Pattern Scout suggestions become tested rules through the rule editor.
- Long-window indicators for slow-drain and collusion patterns.

PRESENTER NOTES

5-minute talk track and demo click path

Time	Say	Click
0:00 – 0:30	Analysts drown in alerts they cannot explain. FinShield links banking, lending and insurance records, flags what is unusual, and shows the evidence for every score and decision.	Overview
0:30 – 1:00	Data in: three formats, validation, rejected rows kept with reasons, everything linked by IDs.	Data sources → stored records, ingestion history
1:00 – 2:00	Explainable risk: a takeover chain scores High 85; a family sharing a tablet stays Low 18 because the household mitigation applies. Same engine, and it shows why.	Overview → T-C002-ATO; Include Low → DEV-HOME
2:00 – 3:15	Level 2: the queue is sorted by priority with factors. The C011 loan ring is one investigation: why grouped, priority, actions (hold is only a recommendation), split, report download.	Cases → CASE-7085d89848 → Why grouped → actions → Download
3:15 – 4:00	Cross-system links: shared device and phone, payout account, mule accounts, insurance claims. Every link is a matching ID; click for records.	View network → click DEV-RING / A-C011
4:00 – 4:40	AI that suggests, never decides: Ask FinShield answers with checked citations, can be saved and rejected; Pattern Scout finds what no rule caught.	Ask FinShield; Network → Run Pattern Scout
4:40 – 5:00	Results: 93% of planted fraud caught at Medium+, 0% of lookalikes at High, on held-out synthetic data. Explainable first, auditable always.	(Evaluation slide)

Before the demo: delete the old local database, start the backend and frontend, click "Load demo dataset". If the network is slow, the AI falls back to quoted evidence, so say "AI off" and continue.